

```
# Conditional statment
#1. if else statment Assignment
number=int(input("Enter the number"))
if number % 2==0:
    print(" The number is Even")
else:
    print(" The number is odd")
```

Enter the number5
The number is odd

```
marks=float(input("Enter the marks "))
if marks >=40:
    print("Pass")
else:
    print("Fail")
```

Enter the marks 87
Pass

```
# if-elif-else statment
number=float(input("Enter the number"))
if number>0:
    print("Positive")
elif number<0:
    print("Negative")
else:
    print("Zero")
```

Enter the number-8
Negative

```
age=int(input("Enter the age "))
if age<0:
    print("Invalid")
elif age<=12:
    print("Child")
elif age<=19:
    print("Teenage")
elif age<=59:
    print("Adult")
else:
    print("senior")
```

Enter the age 13
Teenage

```
#Nested if-else
number =int(input("Enter the number"))
if number>0:
    if number % 2==0:
        print("Positive and Even")
    else:
        print("Positive or odd")
else:
    print("Number is not positive")
```

Enter the number6
Positive and Even

```
age=int(input("Enter the age "))
logged_in=input("Enter the login status true /false ")
if age>18:
    if logged_in=="true":
        print("Access")
    else:
        print("denied")
else:
    print("invalid must be 18")
```

```
Enter the age 90
Enter the login status true /false true
Access
```

```
# ternary operator
marks=int(input("Enter the marks "))
result="pass" if marks>=35 else"fail"
print(result)
```

```
Enter the marks 23
fail
```

```
a=int(input("Enter the number a "))
b=int(input("Enter the number b "))
larger=a if a>b else b
print(larger)
```

```
Enter the number a 7
Enter the number b 45
45
```

```
# Match-case
day=int(input("Enter the day "))
match day:
    case 1:
        print("Today is Monday")
    case 2:
        print("Today is Tuesday")
    case 3:
        print("Today is Wednesday")
    case 4:
        print("Today is Thursday")
    case 5:
        print("Today is Friday")
    case 6:
        print("Today is Saturday")
    case 7:
        print("Today is Sunday")
    case _:
        print("Invalid")
```

```
Enter the day 5
Today is Friday
```

```
grade=input("Enter the grade (A,B,C)")
match grade:
    case "A":
        print("Excellent")
    case "B":
        print("good")
    case "C":
        print("Average")
    case _:
        print("Invalid")
```

```
Enter the grade (A,B,C)A
Excellent
```

```
# Section-2 loops
# for loop
num=int(input("Enter the number "))
for i in range(1,11):
    print(num,"*",i,"=",num*i)
```

```
Enter the number 3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

```
sum=0
for i in range(1,51):
    sum=sum+i
print(sum)
```

```
1275
```

```
#iterating by index sequence
fruits=["Mango","apple","banana","watermelon"]
for i in range(len(fruits)):
    print(i,fruits[i])
```

```
0 Mango
1 apple
2 banana
3 watermelon
```

```
numbers=[12,-4,23,-1,5]
for i in range(len(numbers)):
    if numbers[i]<0:
        numbers[i]=0
print(numbers)
```

```
[12, 0, 23, 0, 5]
```

```
# while loop
num=1
while num<=20:
    if num % 2==0:
        print(num)
    num=num+1
```

```
2
4
6
8
10
12
14
16
18
20
```

```
correct_pass="Admin@123"
password=input("Enter the correct pass")
while password!=correct_pass:
    print("Incorrect pass")
    password=input("Enter the correct pass")
else:
    print("correct pass")
```

```
Enter the correct passAdmin212
Incorrect pass
Enter the correct passAdmin@123
correct pass
```

```
#infinite while loop
while True:
    value=input("Enter the number ( type 'exit' to stop )")
    if value == "exit":
        break
    print(value)
    print("stop")
```

```
Enter the number ( type 'exit' to stop )3
Enter the number ( type 'exit' to stop )6
Enter the number ( type 'exit' to stop )exit
exit
stop
```

```
while True:
    print(" 1. MENU")
    print(" 2. Add")
    print(" 3.quit")
    choice=input("enter your choice ")
    if choice == "1":
        print("Enter the menu")
    elif choice=="2":
        print("add option select")
    elif choice=="3":
        print(" Exit the program ")
        break
    else:
        print("invalid choice")
```

```
1. MENU
2. Add
3.quit
enter your choice 2
add option select
1. MENU
2. Add
3.quit
enter your choice 1
Enter the menu
1. MENU
2. Add
3.quit
enter your choice 3
quit option select
```

```
#Nested loops
n=int(input("Enter the number "))
for i in range(1,n+1):
    for j in range(1,i+1):
```

```
print("",end=" ")
print()
```

```
Enter the number 5
*
* *
* * *
* * * *
* * * * *
```

```
for i in range(1,6):
    for j in range(1,6):
        print(i*j ,end="\t")
    print()
```

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

```
# section 3 loop control statment
# break
numbers=[10,20,30,40,50]
target=int(input("Enter the number "))
for num in numbers:
    if num==target:
        print("found")
        break
else:
    print("not found")
```

```
Enter the number 50
found
```

```
for number in range(1,21):
    print(number)
    if number %7==0:
        break
```

```
1
2
3
4
5
6
7
```

```
# continue
for number in range(1,16):
    if number %3==0:
        continue
    print(number)
```

```
1
2
4
5
7
8
10
11
13
14
```

```
names=["Rohit","Jitendra","Hari","Rahul","akash"]  
for name in names:  
    if name[0] in "aeiou":  
        continue  
    print(name)
```

```
Rohit  
Jitendra  
Hari  
Rahul
```